

Data Flow Diagram

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	2 Marks

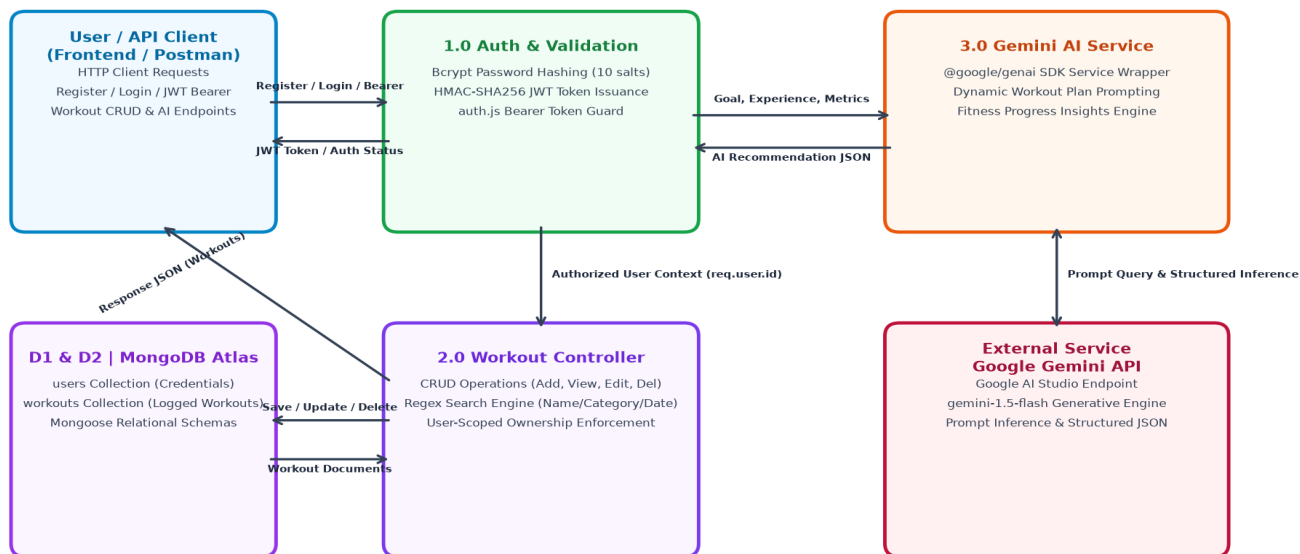
Data Flow Diagram (DFD) Specification:

A Data Flow Diagram (DFD) maps out the flow of information through AI FitTrack API, detailing interactions between external entities, backend processes, and persistent data stores. Use the standard symbol legend below as reference.

Symbol Shape	Element Name	Description	Project Implementation in AI FitTrack API
Oval / Rounded	External Entity	A person, system, or third-party cloud service that sends or receives data.	• User / API Client (Postman Collection / Frontend Web UI) • Google Gemini Generative AI Cloud API
Rectangle with Number	Process	An activity or business logic that transforms incoming data into outgoing data.	• 1.0 Auth & Validation (Bcrypt / JWT / auth.js) • 2.0 Workout Controller (CRUD & Regex Search) • 3.0 Gemini AI Service (Plan Generator & Insights)
Rectangle (Solid / Open)	Data Store	A persistent repository where structured data is stored for later use.	• D1: MongoDB Users Collection (Credentials & Names) • D2: MongoDB Workouts Collection (Logged Activities)
Labeled Arrow	Data Flow	Directed movement of data labeled with specific data fields.	• Credentials, JWT tokens, Workout payloads (duration, calories), Prompt strings, AI Workout Recommendations

Level-1 Data Flow Diagram (DFD) — AI FitTrack API

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Data Flow Process Descriptions:

Step	Source -> Target	Data Transferred	Transformation & Security Logic
1	Client -> Auth Process	name, email, password	Express router validates format; bcrypt hashes password (salt=10); user saved to D1.
2	Auth -> Client	HMAC-SHA256 JWT Token	Payload contains user.id; signed with JWT_SECRET; valid for authorized requests.
3	Client -> Workout Process	Bearer Token + workout data	auth.js verifies token; workoutName, category, duration, calories validated.
4	Workout -> MongoDB (D2)	Workout document object	Persists workoutName, category, duration, calories, workoutDate, user ObjectId.
5	Client -> AI Service	Bearer Token + fitness goals	auth.js verifies token; geminiService constructs prompt with age, goal, experience.
6	AI Service -> Gemini API	Engineered Prompt string	Dispatches request to gemini-1.5-flash endpoint; returns structured recommendation JSON.